

# Experiment 5

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## 1 AIM

Op-Amp Applications:

- 1) Custom Weighted Summing and Difference Amplifier
- 2) Op-Amp Integrator
- 3) Precision Rectifier(Super Diode)

## 2 MATERIALS AND APPARATUS REQUIRED

- 1) Op-Amp (e.g., LM386, LM58)
- 2) Resistors(Carefully selected for weighting)
- 3) Capacitors
- 4) Diode(e.g., 1N4148)
- 5) Function Generator
- 6) DC Power Supply
- 7) Oscilloscope

## 3 THEORY

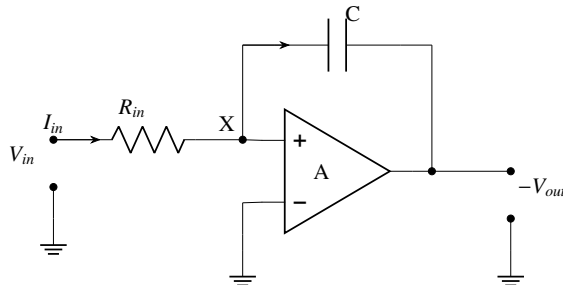
### 3.1 Op-Amp Integrator

#### Circuit Description

The basic configuration of an op-amp integrator consists of:

- An operational amplifier.
- A resistor ( $R_{in}$ ) connected to the inverting input.
- A capacitor ( $C$ ) connected between the output and the inverting input.
- The non-inverting input is grounded.

The circuit diagram is shown below:



### Operation of the Integrator Circuit

The operation of the integrator is based on the virtual ground principle at the inverting input of the op-amp. Assuming an ideal op-amp:

- 1) The input current through  $R_{in}$  is given by:

$$I_{in} = \frac{V_{in}}{R_{in}}.$$

- 2) Since no current flows into the op-amp's inverting terminal, this current flows through the capacitor, causing it to charge or discharge.
- 3) The voltage across the capacitor ( $V_C$ ) is related to the charge ( $Q$ ) by:

$$Q = CV_C.$$

The current through the capacitor is:

$$I_C = C \frac{dV_C}{dt}.$$

Equating  $I_{in}$  and  $I_C$ , we have:

$$\frac{V_{in}}{R_{in}} = C \frac{dV_{out}}{dt}.$$

Rearranging and integrating both sides with respect to time gives:

$$V_{out} = -\frac{1}{R_{in}C} \int V_{in} dt,$$

where the negative sign indicates that the output is inverted.

Thus, the output voltage ( $V_{out}$ ) is proportional to the time integral of the input voltage ( $V_{in}$ ).

### 3.2 Half and Full-Wave Rectifiers

#### Half-Wave Precision Rectifier

The half-wave precision rectifier allows only one half of the input AC signal (positive or negative) to pass through while blocking the other half. This is achieved using an op-amp and a diode configured such that the op-amp compensates for the diode's forward voltage drop.

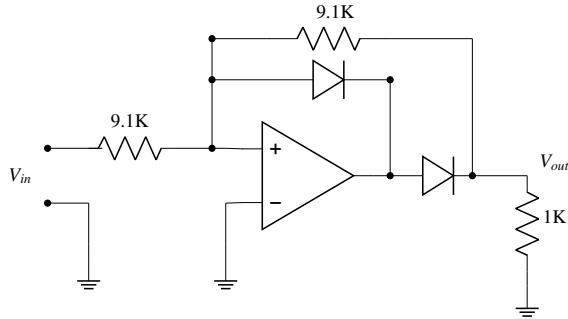
*Circuit Description:*

- 1) An operational amplifier (e.g., LM358).
- 2) A diode (e.g., 1N4148) connected in feedback.
- 3) Resistors to set the input and feedback paths.

*Working Principle:* When the input signal ( $V_{in}$ ) is positive, the op-amp output drives the diode into conduction, and the output ( $V_{out}$ ) follows  $V_{in}$ . When  $V_{in}$  is negative, the diode becomes reverse-biased, blocking current flow, and  $V_{out}$  remains at zero.

Mathematically:

$$V_{out} = \begin{cases} V_{in}, & V_{in} > 0, \\ 0, & V_{in} \leq 0. \end{cases}$$



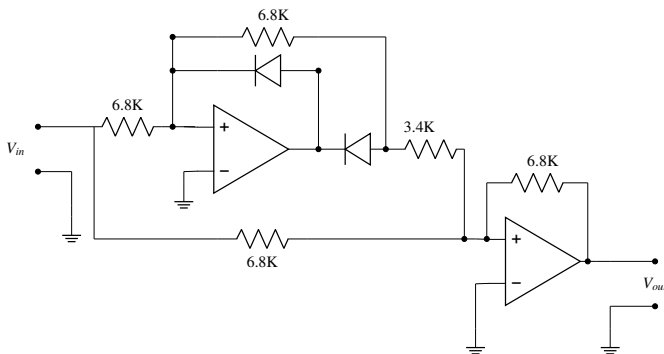
This results in a rectified waveform where only the positive half-cycles are present at the output.

### *Full-Wave Precision Rectifier*

The full-wave precision rectifier allows both halves of the input AC signal to be rectified, producing a unipolar output. This is achieved by combining a half-wave rectifier with a summing amplifier.

#### *Circuit Description:*

- 1) A half-wave precision rectifier circuit.
- 2) A summing amplifier to combine signals from both positive and negative cycles.
- 3) Additional diodes and resistors for proper signal routing.



#### *Working Principle:*

- 1) During the positive half-cycle of  $V_{in}$ , the first op-amp operates as a half-wave rectifier, passing the positive signal to the summing amplifier.
- 2) During the negative half-cycle of  $V_{in}$ , another path inverts and shifts the signal such that it also appears as a positive contribution at the summing amplifier input. The summing amplifier combines these contributions to produce a continuous positive output for both halves of the input signal.

Mathematically: For an input sinusoidal signal  $V_{in} = V_m \sin(\omega t)$ , the output voltage is:

$$V_{out} = |V_{in}| = V_m |\sin(\omega t)|.$$

This results in a fully rectified waveform at the output.

## 4 PROCEDURE

### 4.1 Op-Amp Integrator

- 1) Set Up the Circuit:
  - Connect the circuit as per the integrator circuit diagram.
  - Use a resistor ( $R_{in}$ ) at the input and a capacitor ( $C_f$ ) in the feedback loop of the op-amp.
  - Ground the non-inverting terminal of the op-amp.
  - Connect the power supply to the op-amp ( $+V_{cc}$  and  $-V_{cc}$ ).
- 2) Apply Input Signal:
  - Connect a function generator to provide a square wave input signal to the circuit.
  - Set the input signal to  $2 V_{pp}$  amplitude and 1 kHz frequency.
- 3) Observe Output Waveform:
  - Connect the input and output signals to two channels of a CRO.
  - Set the CRO to AC coupling mode.
  - Observe and record both input and output waveforms simultaneously.
- 4) Analyze Output Waveform:
  - For a square wave input, verify that the output is a triangular waveform.
  - Note down the amplitude and time period of both input and output signals.

### 4.2 Precision Rectifiers

- 1) Set Up the Circuit:
  - Connect the circuit as per the precision rectifier circuit diagram.
  - For the half-wave rectifier:
    - Use an op-amp with a diode in the feedback loop and a resistor at the input.
    - Ground the non-inverting terminal of the op-amp.
  - For the full-wave rectifier:
    - Combine two op-amps: one for inverting and rectifying the negative half-cycle and another for summing both positive and inverted signals.
    - Use appropriate diodes and resistors for signal routing.
  - Connect the power supply to the op-amp ( $+V_{cc}$  and  $-V_{cc}$ ).
- 2) Apply Input Signal:
  - Connect a function generator to provide a sine wave input signal to the circuit.
  - Set the input signal to  $2 V_{pp}$  amplitude and 1 kHz frequency.
- 3) Observe Output Waveform:
  - Connect the input and output signals to two channels of a CRO.

- Set the CRO to AC coupling mode.
- Observe and record both input and output waveforms simultaneously.

4) Analyze Output Waveform:

- For the half-wave rectifier:
  - Verify that only one half (positive or negative) of the sine wave is present at the output while the other half is blocked.
  - Note down the amplitude and time period of both input and output signals.
- For the full-wave rectifier:
  - Verify that both halves of the sine wave are rectified, resulting in a unipolar output waveform.
  - Note down the amplitude and time period of both input and output signals.

## 5 OBSERVATIONS

### 5.1 Op-Amp Integrator

## 6 PRECAUTIONS

- 1) Make sure not to accidentally discharge the capacitor
- 2) Make sure the DC generator is connected to ground properly
- 3) Use a push button for shorting the inductor and capacitor
- 4) Repeat the experiment multiple times to be sure of the reading as it is a sensitive measurement

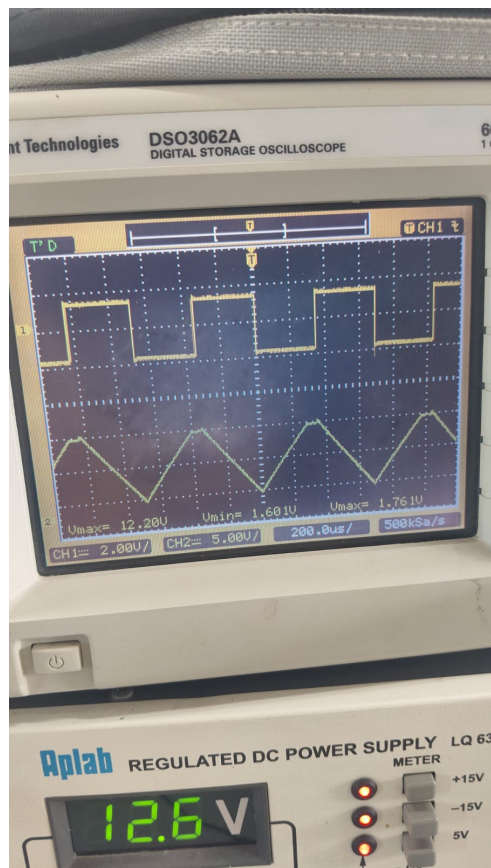


Fig. 1: Op-Amp Integrator with input 1452 Hz

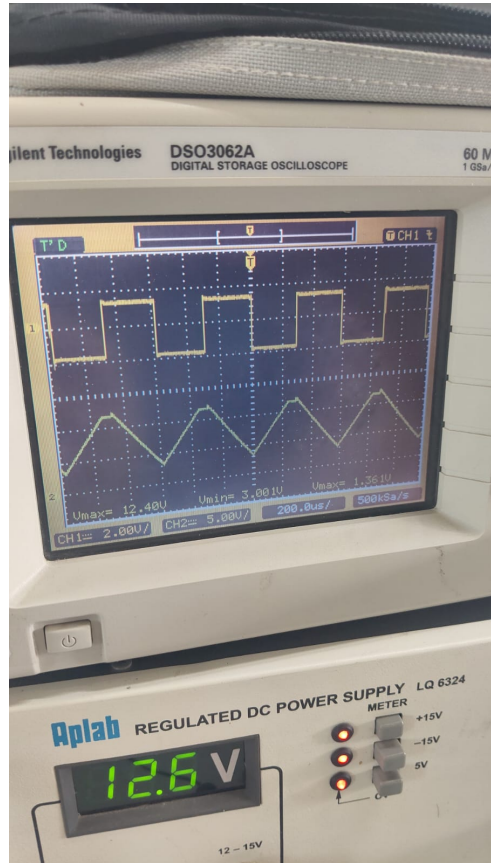


Fig. 2: Op-Amp Integrator with input 1708 Hz

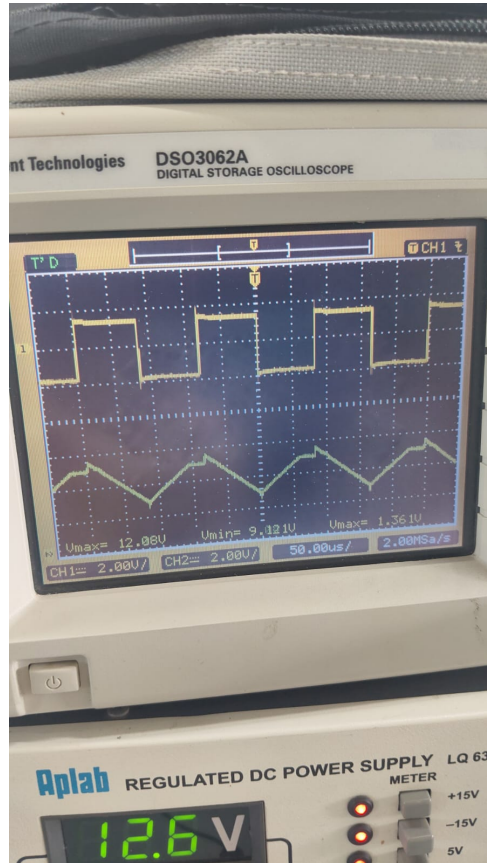


Fig. 3: Op-Amp Integrator with input 6049 Hz